



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

COS 710: Artificial Intelligence

Assignment 3: Structure-Based Grammatical Evolution for Regression

Due Date: 24 May, 23:30

This assignment involves implementing structure-based grammatical evolution to predict the electricity load for a specific time on a particular day given the data for the n previous days at the same time or alternatively the m previous values can be used. The problem is a time series problem. The dataset can be accessed from <https://drive.google.com/drive/folders/1q9JxS7Gy-Fyk0dAl6zWEquMjl1F0fhXM?usp=sharing>.

Assignments must be submitted via clickUP. The source code, compiled code and report must be submitted. Please note that the assignment can only be coded in Java, C++ or Python.

The report must include:

- Description of the data set .
- Description of the performance metric used to assess the quality of the predictor.
- Description of how structure is incorporated into canonical GE.
- Description of the grammar used.
- Description of initial population generation.
- Description of the fitness function and fitness evaluation.
- Description of the selection method.
- Description of genetic operators

- A description of the experimental setup, i.e. parameter values used, technical specifications of the machine used to develop the program and run simulations.
- The results (averages, standard deviation and best value over the runs performed) and a discussion of the results. A minimum of 10 runs should be performed.
- Runtimes of the algorithm.
- Comparison of the results with Assignment 1 and Assignment 2.

Total: 50